

PHASMOPHOBIA

A ghost hunting game built for augmented reality

JENNIFER WAGNER · SIMON WINTER · GYURI LEE · FANGXING LIU



One hidden ghost, zero traditional input

You are a Ghost hunter in your own house and try your best to survive.

Scan your physical space to uncover a supernatural entity hiding in plain sight.

Deploy printed marker cards in the real-world to spawn 3D ghost-hunting tools exactly where you need them.

Gather evidence, identify the spirit and capture it before it disappears, or before it gets you.



Device movement

6-DoF camera pose is the controller; for gathering evidence, revealing and capturing



Printed marker cards

Physical placement of virtual ghost-hunting tools via image tracking



Tablet as Ghost Hunting Tool

Scanner modes, journal, ghost-type confirmation



Gameplay Loop

- 1 **SCAN** the room until tracking is stable
- 2 **INVESTIGATE** and collect evidence with AR Marker tools
- 3 **IDENTIFY** the ghost in the journal
- 4 **CAPTURE** the ghost
- 5 **CLOSE** the case

2.5 GET JUMPSCARED



AR Features



Spatial Understanding

- ▶ Detects **surfaces** and **feature points** in the room.
- ▶ Generates **mesh** and **depth data** when the device supports it.



Ghost Placement

- ▶ Detected planes, feature points, and room bounds dictate a believable **spawn position**.
- ▶ Keeps the ghost rigidly fixed in its real-world location.



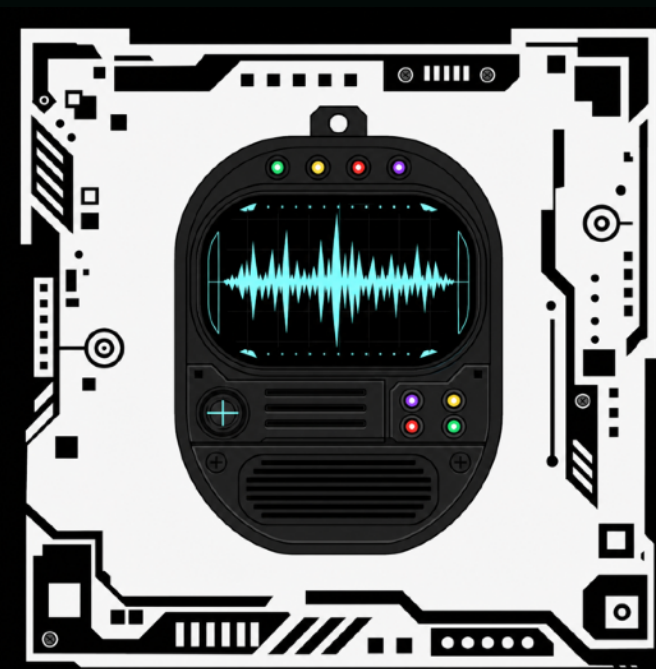
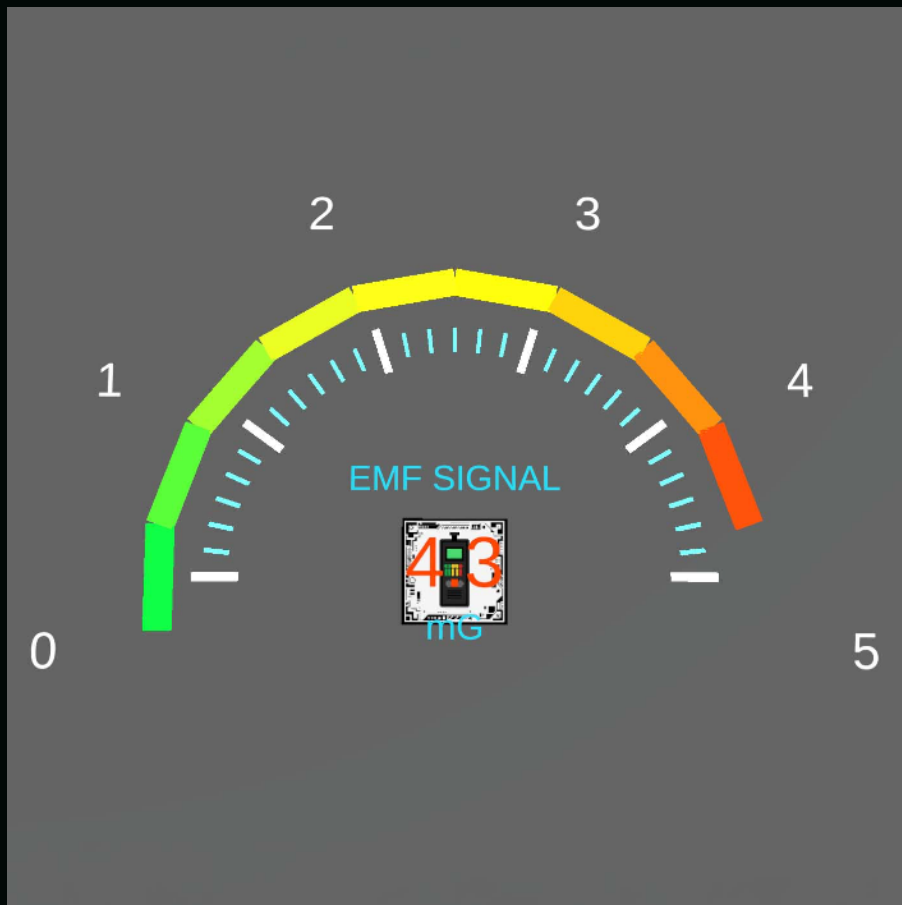
Physical Tool Cards

- ▶ Recognizes **printed marker cards**.
- ▶ **Moves** the matching AR tool prefab **alongside** the physical card.

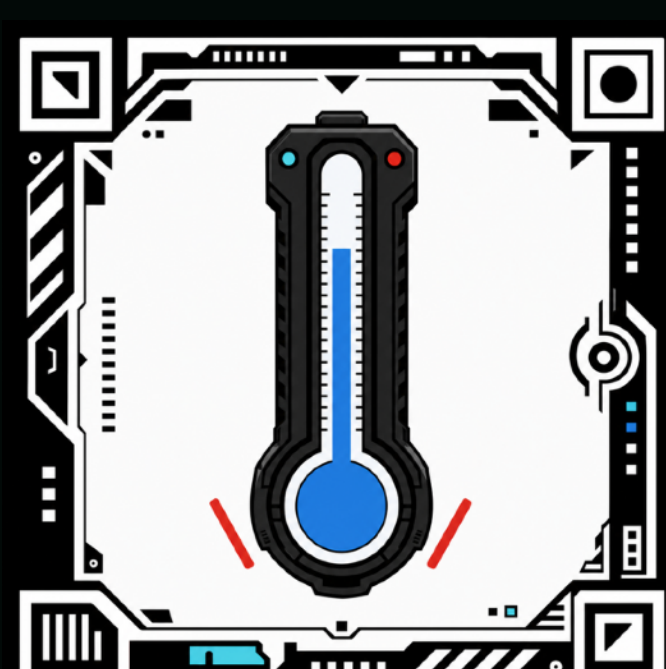
Physical Marker Tools



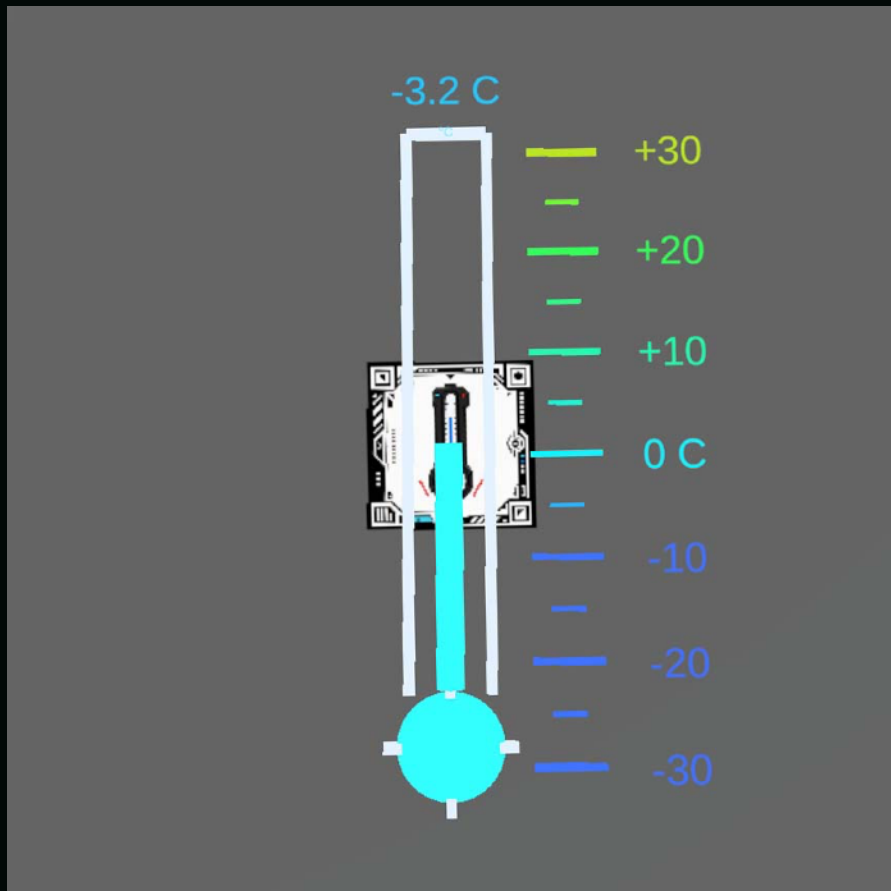
EMF Reader

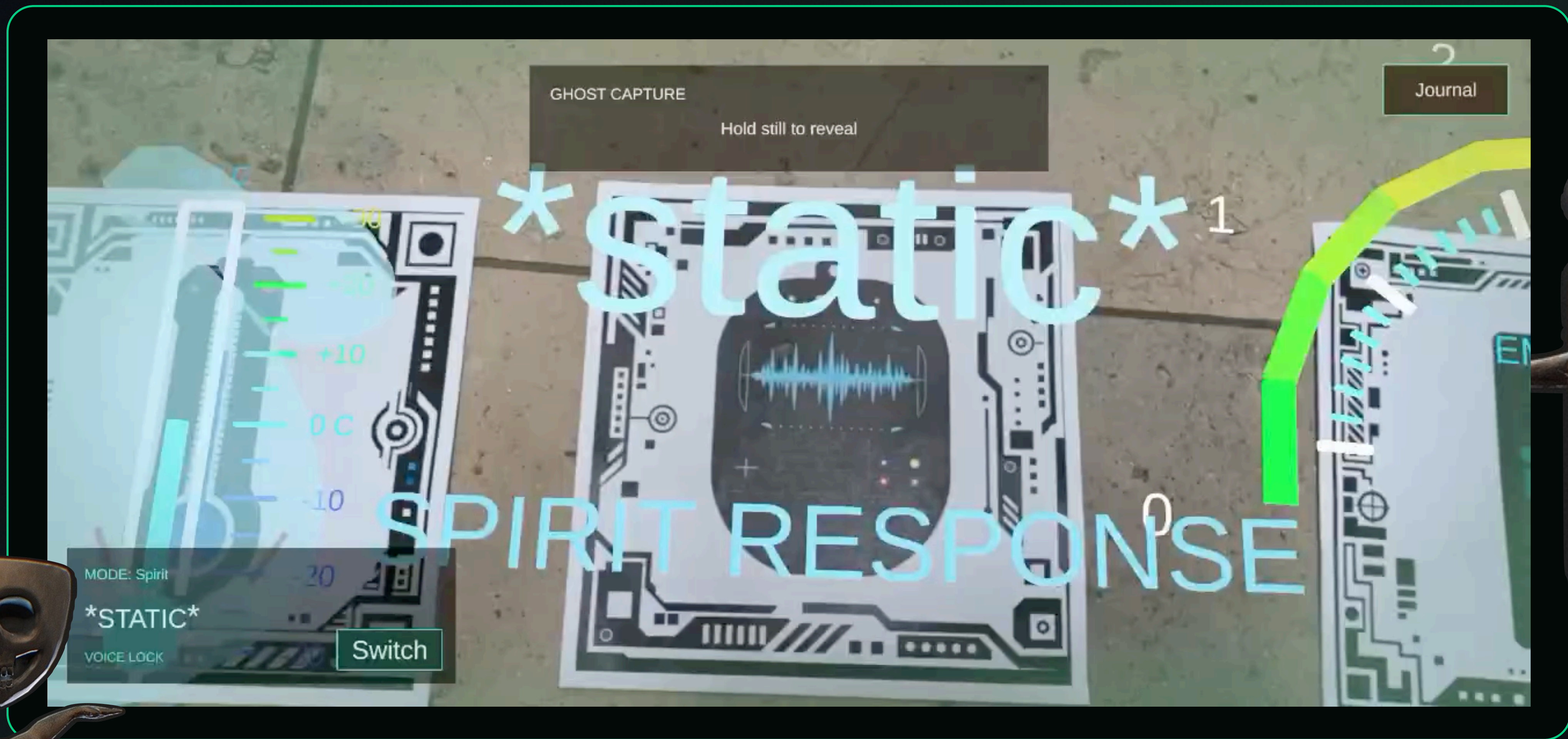
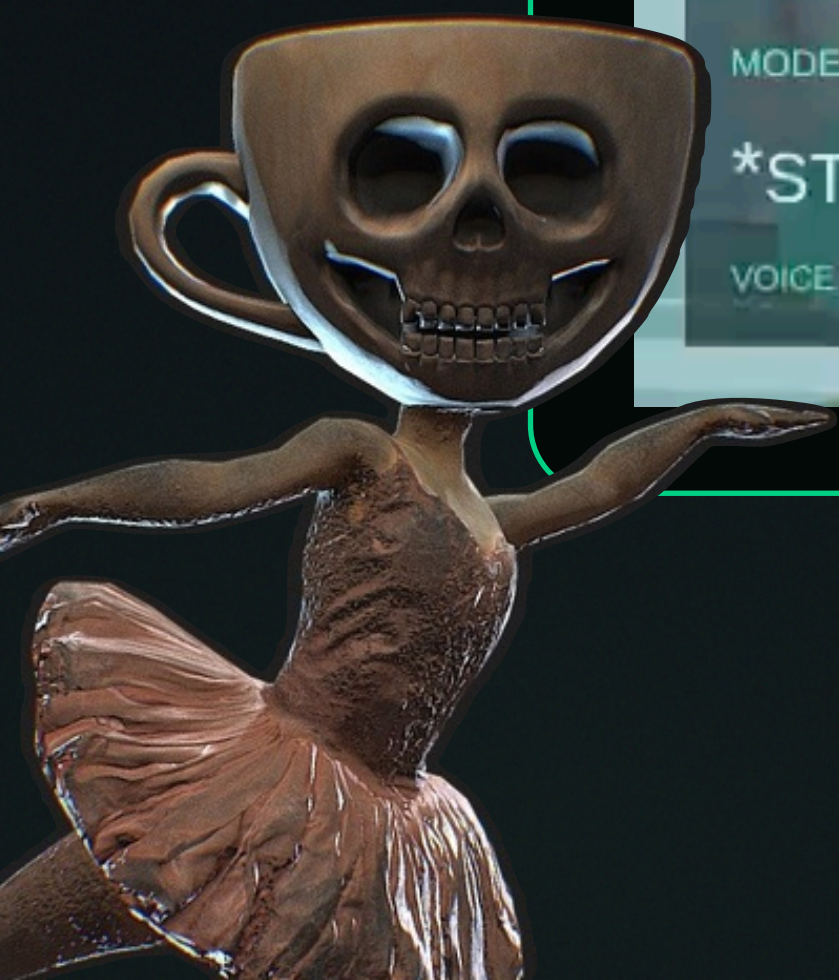


Spirit Box



Thermometer





Marker Scanning Challenges



Marker Detection

Challenge:

- AR marker scanning failed frequently or yielded poor tracking accuracy in environments with **poor lighting** conditions.

Solution:

- AR marker scanning failed frequently or yielded poor tracking accuracy in environments with **poor lighting** conditions.

Tracking Loss Persistence

Challenge:

- Spawning/virtual tools were not removed properly when the tracking marker exited the field of view, remaining **stuck at the last known position**.

Solution:

- Managed different **Unity AR tracking states** dynamically, combined with **removal timeouts** to safely clear objects upon loss of marker view.

🎯 AR Ghost Spawning Challenges



Challenge:

- Creating ghost spawning that felt believable in AR was difficult because the ghost had to appear in a **real, unpredictable room** while still supporting gameplay tension
- A spawn point could not be too close, too far away, floating incorrectly, or not be found at all

Solution:

- Designed the spawning logic around **spatial constraints**
- Ghost is placed at a **controlled distance** from the player, aligned with the detected environment, and positioned to preserve visibility and surprise
- User can **retry scanning** when no spawn points can be found

🎯 Create feeling of suspense



Challenge:

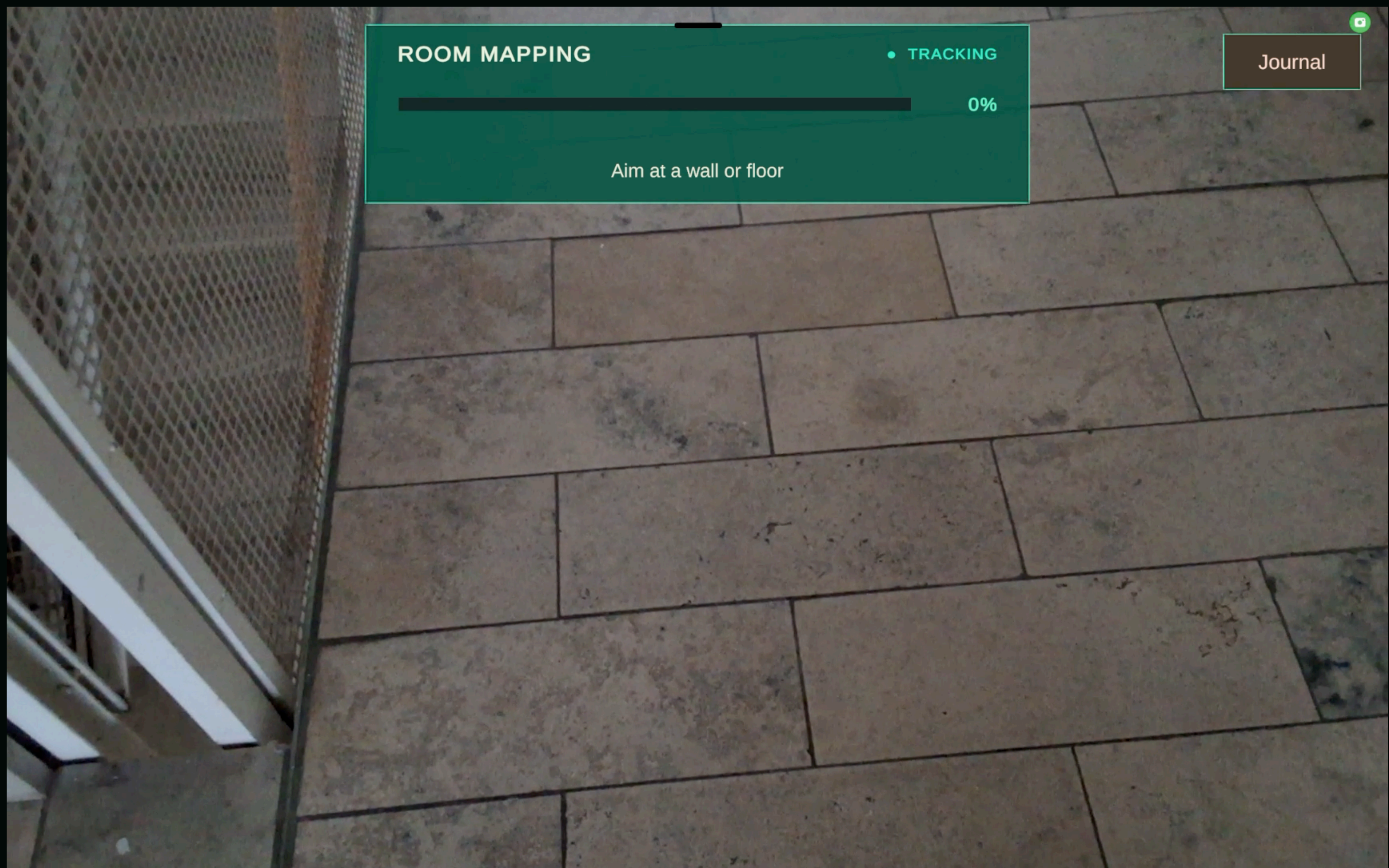
- Fear in AR is hard to create because the player is in a real environment, so the scare must **feel believable**
- If ghost appearances are too predictable, too frequent, or poorly timed, the player quickly **stops feeling tension**

Solution:

- Use controlled unpredictability with delayed appearances, **sudden ghost jump scares**, and text to speech
- Combine believable ghost placement with **spatial audio** and **visual cues**

1

SCAN the room until tracking is stable



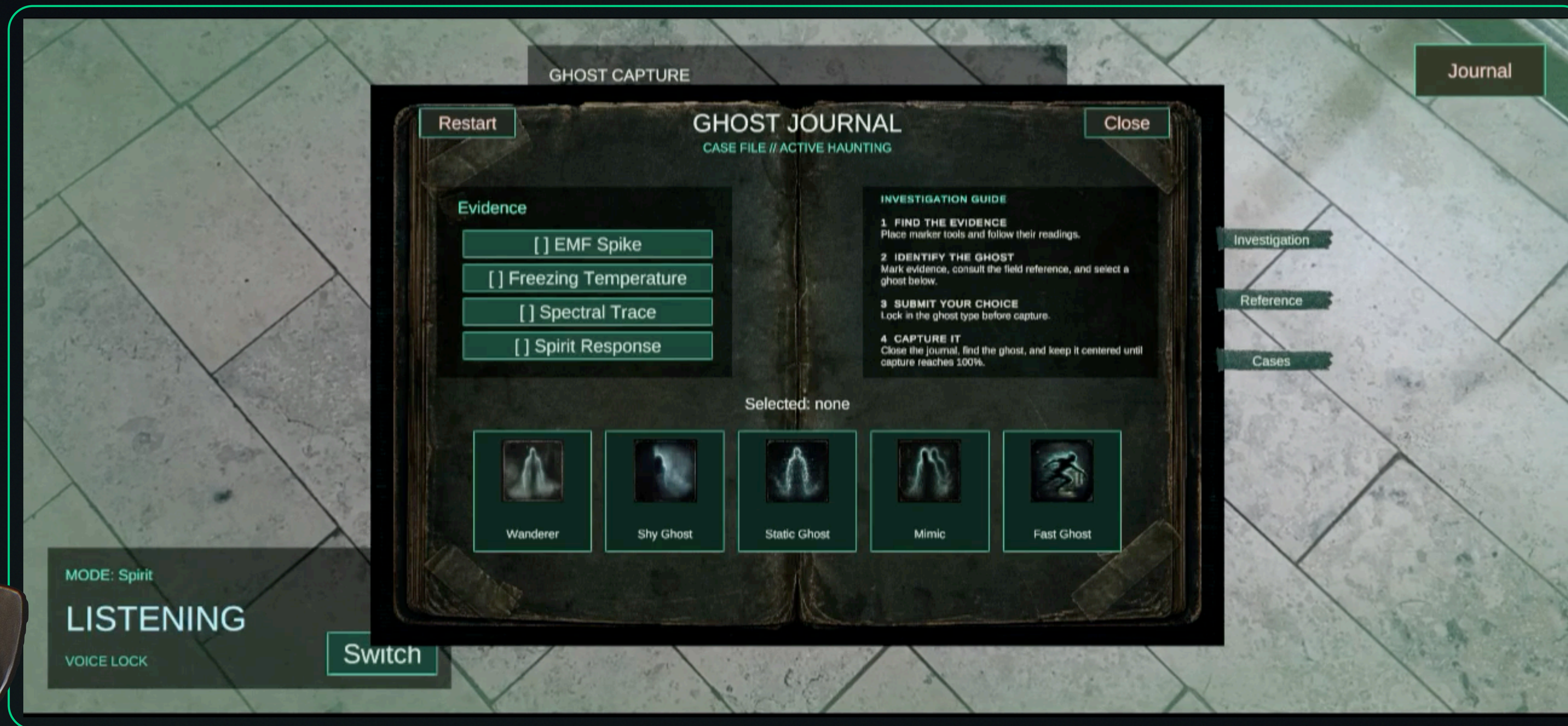
2

INVESTIGATE and collect evidence with AR Marker tools



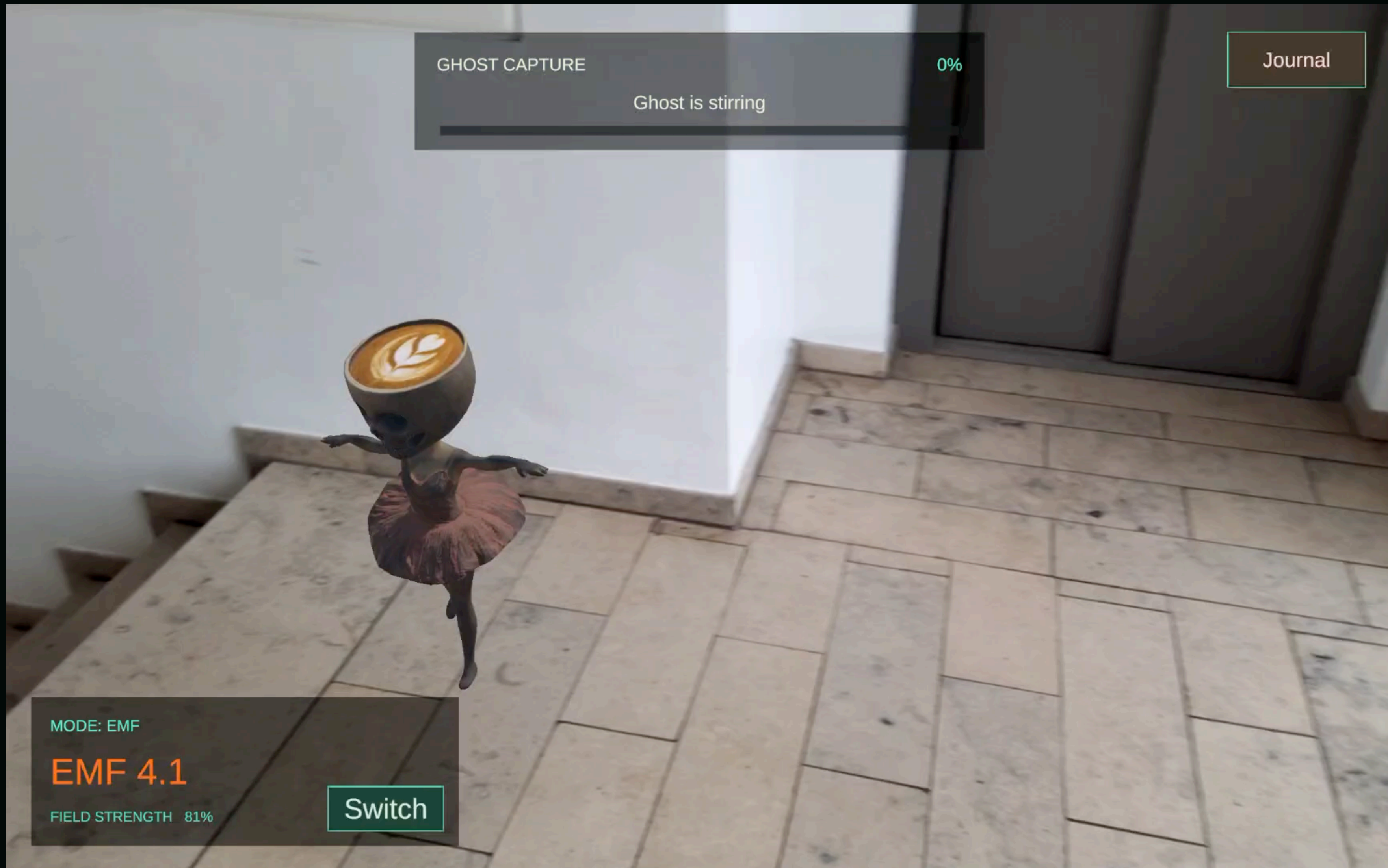
3

IDENTIFY the ghost in the journal



4

CAPTURE the ghost



5

CLOSE the case

CASE REPORT

CASE SOLVED

Your identification matched the haunting.

Entity: Wanderer

Filed as: Wanderer

Capture: Success (100% in 3.9s)

Evidence marked: EMF Spike, Spectral Trace

NEXT ROUND



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Thank you for listening!
Any questions?

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